

B1 Fluids — Model Solutions, Trinity 2024

Throughout, the Navier–Stokes equation for an incompressible viscous fluid under gravity is

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} + \frac{1}{\rho} \nabla p + g \hat{\mathbf{z}} = \nu \nabla^2 \mathbf{v}, \quad (1)$$

with ν the kinematic viscosity. We also use $1000 \text{ mb} = 10^5 \text{ Pa}$.

Question 1 — Stream functions and spin-down of a tank

Problem. Define streamfunction and vorticity for 2D incompressible flow; verify $\psi = -\Omega(x^2 + y^2)/2$ is solid-body rotation. In an elliptical tank $(x/L_x)^2 + (y/L_y)^2 = 1$ with the turntable spinning, sketch laboratory streamlines. After the turntable is halted abruptly (inviscid bulk), find the vorticity, verify $\psi = Ax^2 + By^2$, determine A, B , sketch streamlines, and discuss paddle-wheel rotation in the bulk and near the wall at high Re .

(a) Stream function, vorticity, and solid body rotation

For 2D incompressible flow $\mathbf{v} = (v_x, v_y, 0)$, continuity $\partial_x v_x + \partial_y v_y = 0$ admits a streamfunction:

$$v_x = \frac{\partial \psi}{\partial y}, \quad v_y = -\frac{\partial \psi}{\partial x}. \quad (2)$$

$\mathbf{v} \cdot \nabla \psi = 0$, so $\psi = \text{const}$ are streamlines. Vorticity in 2D:

$$\omega = \frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y}. \quad (3)$$

Substitute (2):

$$\omega = -\nabla^2 \psi. \quad (4)$$

For $\psi = -\frac{1}{2}\Omega(x^2 + y^2)$, apply (2):

$$v_x = -\Omega y, \quad v_y = \Omega x.$$

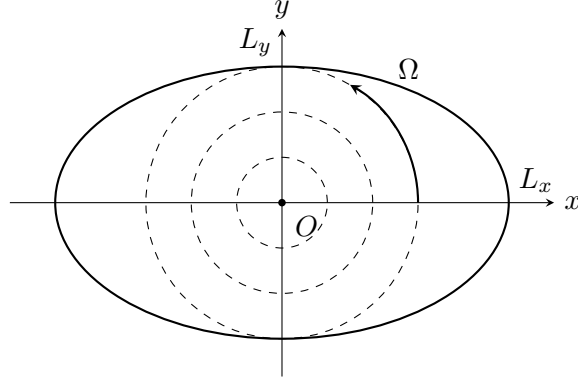
In polar coordinates: $\mathbf{v} = \Omega r \hat{\boldsymbol{\theta}}$, i.e. rigid rotation at rate Ω . Apply (4):

$$\omega = -\nabla^2 \left[-\frac{1}{2}\Omega(x^2 + y^2) \right] = 2\Omega.$$

$$\boxed{\omega = 2\Omega.} \quad (5)$$

(b) Streamlines in the laboratory frame for an elliptical tank

In the laboratory frame the spun-up fluid rotates rigidly at Ω , so streamlines are concentric circles centred on the tank axis.



The circular streamlines cross the elliptical wall near the major-axis tips ($r \approx L_x$). No fluid actually passes through the wall: the wall is co-rotating, and at each instant it occupies the position the streamline indicates. Steady streamlines coincide with a rigid boundary only in the frame in which that boundary is at rest — here, the rotating frame, where the fluid is at rest and no streamlines exist.

(c) Vorticity after the turntable is halted; verification of $\psi = Ax^2 + By^2$

Inviscid bulk: Kelvin's theorem freezes vorticity onto fluid elements. Just before stopping, every element had $\omega = 2\Omega$; just after,

$$\omega = 2\Omega. \quad (6)$$

For $\psi = Ax^2 + By^2$, apply (2):

$$v_x = 2By, \quad v_y = -2Ax.$$

Apply (4):

$$\omega = -2(A + B).$$

Match with (6):

$$A + B = -\Omega. \quad (i)$$

Wall is a streamline, so $\psi = \text{const}$ on $(x/L_x)^2 + (y/L_y)^2 = 1$:

$$A = \frac{C}{L_x^2}, \quad B = \frac{C}{L_y^2}. \quad (ii)$$

Substitute (ii) into (i):

$$C \left(\frac{1}{L_x^2} + \frac{1}{L_y^2} \right) = -\Omega.$$

Solve for C :

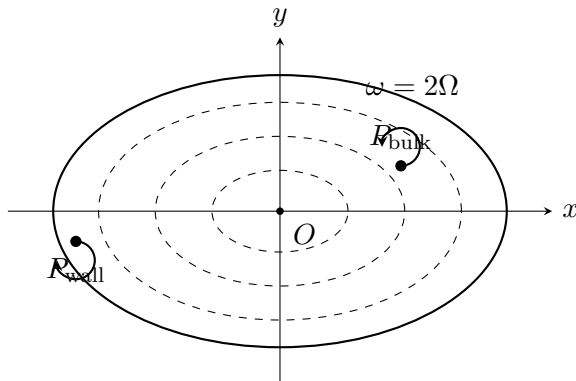
$$C = -\frac{\Omega L_x^2 L_y^2}{L_x^2 + L_y^2}.$$

Substitute back:

$$\boxed{A = -\frac{\Omega L_y^2}{L_x^2 + L_y^2}, \quad B = -\frac{\Omega L_x^2}{L_x^2 + L_y^2}.} \quad (7)$$

(d) Streamlines in the spin-down flow and paddle wheel motion

Streamlines $\psi = \text{const}$ are nested ellipses similar to the wall, $(x/L_x)^2 + (y/L_y)^2 = \text{const}$.



Bulk paddle wheel: local vorticity $\omega = 2\Omega$ equals twice the local angular velocity, so the wheel spins on its own axis at rate Ω in the original sense while drifting along an ellipse.

Boundary layer: at high Re viscosity has only diffused a sliver of thickness $\delta \sim \sqrt{\nu t}$ into the wall. No-slip pulls the tangential velocity from its inviscid value down to zero across δ :

$$\omega_{\text{wall}} \sim -\frac{v_t}{\delta},$$

of opposite sign to the bulk 2Ω . A wheel sitting in this layer therefore spins counter to one in the bulk.

Question 2 — Velocity potential, Bernoulli, deep-water surface waves

Problem. Define the velocity potential, derive Laplace's equation and the unsteady Bernoulli equation. For deep-water waves $\eta = A \cos(kx - \omega t)$, derive the linearised kinematic boundary condition, the dynamic surface BC, solve for ϕ , derive $\omega = \sqrt{gk}$, compare phase and group speeds; sketch a wave packet. Find v_x, v_z at the surface, sketch the parcel orbit, and state the vorticity.

(a) Potential, Laplace's equation and the Bernoulli relation

Irrotational flow $\nabla \times \mathbf{v} = 0$:

$$\mathbf{v} = \nabla \phi. \quad (8)$$

Impose incompressibility $\nabla \cdot \mathbf{v} = 0$:

$$\nabla^2 \phi = 0. \quad (9)$$

Inviscid Navier–Stokes (1) with $\nu = 0$, $\rho = \text{const}$. Vector identity for irrotational flow:

$$(\mathbf{v} \cdot \nabla) \mathbf{v} = \frac{1}{2} \nabla |\mathbf{v}|^2 - \mathbf{v} \times (\nabla \times \mathbf{v}) = \frac{1}{2} \nabla |\nabla \phi|^2.$$

Substitute (8) and use $g\hat{\mathbf{z}} = \nabla(gz)$:

$$\nabla \left(\frac{\partial \phi}{\partial t} + \frac{1}{2} |\nabla \phi|^2 + \frac{p}{\rho} + gz \right) = 0.$$

Integrate (absorb time-dependent constant into ϕ):

$$\boxed{\frac{\partial \phi}{\partial t} + \frac{|\nabla \phi|^2}{2} + \frac{p}{\rho} + gz = C.} \quad (10)$$

Terms: $\partial_t \phi$ unsteady, $\frac{1}{2}|\mathbf{v}|^2$ kinetic, p/ρ pressure work, gz gravity. Holds throughout the irrotational fluid at any instant.

(b) Linearised boundary conditions, dispersion relation, group speed

Kinematic BC $D(z - \eta)/Dt = 0$ at $z = \eta$:

$$\frac{\partial \eta}{\partial t} + v_x \frac{\partial \eta}{\partial x} = v_z \quad \text{at } z = \eta. \quad (11)$$

Linearise (drop $v_x \partial_x \eta$, evaluate at $z = 0$):

$$\left. \frac{\partial \phi}{\partial z} \right|_{z=0} = \frac{\partial \eta}{\partial t}. \quad (12)$$

Insert $\eta = A \cos(kx - \omega t)$:

$$\left. \frac{\partial \phi}{\partial z} \right|_{z=0} = \omega A \sin(kx - \omega t). \quad (13)$$

Dynamic BC: pressure on the surface equals atmospheric (absorb into C). Evaluate (10) at $z = \eta$, drop $\frac{1}{2}|\nabla \phi|^2$ as second order, and Taylor to $z = 0$:

$$\left. \frac{\partial \phi}{\partial t} \right|_{z=0} + g\eta = 0. \quad (14)$$

Deep water: $\phi \rightarrow 0$ as $z \rightarrow -\infty$. Take $\phi = F(z) \sin(kx - \omega t)$ in $\nabla^2 \phi = 0$:

$$F''(z) - k^2 F(z) = 0.$$

Decaying solution ($k > 0$):

$$F(z) = B e^{kz}.$$

Apply (13):

$$kB = \omega A.$$

Solve for B :

$$B = \frac{\omega A}{k}.$$

So

$$\phi = \frac{\omega A}{k} e^{kz} \sin(kx - \omega t). \quad (15)$$

Differentiate in t at $z = 0$:

$$\left. \frac{\partial \phi}{\partial t} \right|_{z=0} = -\frac{\omega^2 A}{k} \cos(kx - \omega t).$$

Substitute into (14):

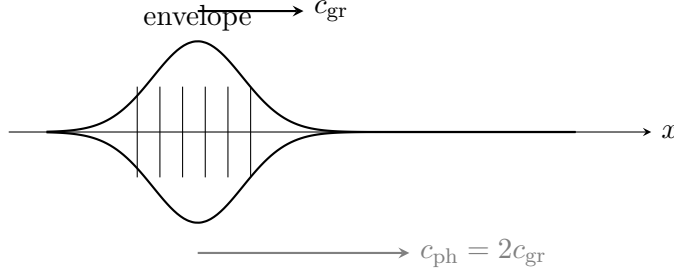
$$-\frac{\omega^2 A}{k} \cos(kx - \omega t) + gA \cos(kx - \omega t) = 0.$$

Divide by $A \cos(kx - \omega t)$:

$$\frac{\omega^2}{k} = g. \quad \boxed{\omega = \sqrt{gk}}. \quad (16)$$

Phase and group speeds from (16):

$$c_{\text{ph}} = \frac{\omega}{k} = \sqrt{\frac{g}{k}}, \quad c_{\text{gr}} = \frac{d\omega}{dk} = \frac{1}{2} \sqrt{\frac{g}{k}} = \frac{1}{2} c_{\text{ph}}.$$



A wave packet is a localised superposition; its envelope (and energy) move at c_{gr} , while individual crests propagate at $c_{ph} = 2c_{gr}$, born at the rear and dying at the front. Deep-water gravity waves are dispersive.

(c) Surface velocity, parcel trajectory and vorticity

From (15), gradient at $z = 0$:

$$v_x = \left. \frac{\partial \phi}{\partial x} \right|_{z=0} = \omega A \cos(kx - \omega t).$$

$$v_z = \left. \frac{\partial \phi}{\partial z} \right|_{z=0} = \omega A \sin(kx - \omega t).$$

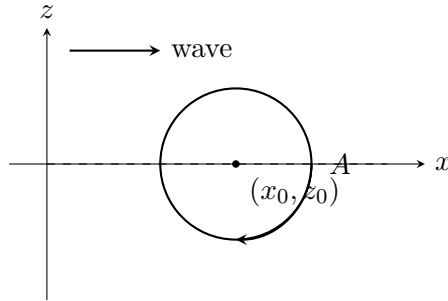
Lagrangian trajectory at leading order: integrate $\dot{x} = v_x(x_0, 0, t)$, $\dot{z} = v_z(x_0, 0, t)$:

$$x(t) - x_0 = -A \sin(kx_0 - \omega t).$$

$$z(t) - z_0 = A \cos(kx_0 - \omega t).$$

Square and add:

$$[x - x_0]^2 + [z - z_0]^2 = A^2.$$



Surface parcel moves on a circle of radius A , traversed clockwise for $+x$ -propagating waves: forward at the crest, downward on the front face, backward in the trough, upward on the rear.

Vorticity in the (x, z) -plane:

$$\omega_y = \frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} = k\omega A \cos(kx - \omega t) - k\omega A \cos(kx - \omega t).$$

Cancel:

$$\boxed{\omega_y = 0.} \tag{17}$$

The flow was assumed irrotational ($\mathbf{v} = \nabla \phi$); each parcel orbits without spinning about its own centre.

Question 3 — Squeeze film between a disk and a table

Problem. A disk of radius R above a smooth table at $z = 0$ is separated by viscous fluid of thickness $h(t) \ll R$, with a small central hole at $r \leq a \ll R$. (a) From (1) derive the lubrication equation, the no-slip BCs on v_r , and show $v_r = -(2\rho\nu)^{-1}\partial_r p z(h - z)$. (b) From continuity derive $v_z(z = h)$ and the pressure profile. (c) Compute the lift force for $a = 0$, $R = 1$ m, $h = 0.5$ mm, $\dot{h} = 1$ mm s⁻¹, $\rho = 10^3$ kg m⁻³, $\nu = 10^{-6}$ m² s⁻¹, and discuss the effect of a hole.

(a) Thin-film derivation and parabolic velocity profile

Cylindrical (r, θ, z) , axisymmetric, ignore gravity. Radial component of (1):

$$\frac{\partial v_r}{\partial t} + v_r \frac{\partial v_r}{\partial r} + v_z \frac{\partial v_r}{\partial z} + \frac{1}{\rho} \frac{\partial p}{\partial r} = \nu \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial v_r}{\partial r} \right) - \frac{v_r}{r^2} + \frac{\partial^2 v_r}{\partial z^2} \right]. \quad (18)$$

Lubrication: $h \ll R$, reduced Reynolds number $\text{Re } h^2/R^2 \ll 1$. Drop inertia, drop radial viscous terms (smaller by $(h/R)^2$):

$$\frac{1}{\rho} \frac{\partial p}{\partial r} = \nu \frac{\partial^2 v_r}{\partial z^2}. \quad (19)$$

Vertical momentum gives $\partial_z p = 0$, so $p = p(r, t)$.

No-slip at table and disk:

$$v_r(z = 0) = 0, \quad v_r(z = h) = 0.$$

Integrate (19) twice in z (treat $\partial_r p$ as z -independent):

$$v_r = \frac{1}{2\rho\nu} \frac{\partial p}{\partial r} z^2 + C_1(r)z + C_2(r).$$

Apply $v_r(0) = 0$:

$$C_2 = 0.$$

Apply $v_r(h) = 0$:

$$C_1 = -\frac{h}{2\rho\nu} \frac{\partial p}{\partial r}.$$

Therefore

$$\boxed{v_r = -\frac{1}{2\rho\nu} \frac{\partial p}{\partial r} z(h - z).} \quad (20)$$

(b) Continuity, vertical velocity and pressure profile

The relation $r^{-1}\partial_r(rv_r) + \partial_z v_z = 0$ is $\nabla \cdot \mathbf{v} = 0$ in cylindrical coordinates for axisymmetric flow.

Integrate continuity from $z' = 0$ ($v_z = 0$ at the table) to z :

$$v_z(r, z, t) = -\int_0^z \frac{1}{r} \frac{\partial}{\partial r} [rv_r] dz'.$$

Substitute (20) and pull the r -only factor out:

$$v_z = \frac{1}{2\rho\nu} \frac{1}{r} \frac{\partial}{\partial r} \left[r \frac{\partial p}{\partial r} \right] \int_0^z z'(h - z') dz'.$$

Evaluate the inner integral:

$$\int_0^z z'(h - z') dz' = \frac{1}{2}hz^2 - \frac{1}{3}z^3.$$

Set $z = h$:

$$v_z(r, h, t) = \frac{h^3}{12\rho\nu} \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial p}{\partial r} \right).$$

Match to disk velocity, $v_z(r, h, t) = \dot{h}$:

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial p}{\partial r} \right) = \frac{12\rho\nu\dot{h}}{h^3}.$$

Integrate once:

$$r \frac{\partial p}{\partial r} = \frac{6\rho\nu\dot{h}}{h^3} r^2 + C_1.$$

Divide by r and integrate again:

$$p(r, t) = \frac{3\rho\nu\dot{h}}{h^3} r^2 + C_1 \ln r + C_2.$$

Pressure BCs: $p(R) = p_0$ at outer rim, $p(a) = p_0$ at hole:

$$p_0 = \frac{3\rho\nu\dot{h}}{h^3} R^2 + C_1 \ln R + C_2.$$

$$p_0 = \frac{3\rho\nu\dot{h}}{h^3} a^2 + C_1 \ln a + C_2.$$

Subtract:

$$0 = \frac{3\rho\nu\dot{h}}{h^3} (R^2 - a^2) + C_1 \ln(R/a).$$

Solve for C_1 :

$$C_1 = \frac{3\rho\nu\dot{h}}{h^3} \frac{R^2 - a^2}{\ln(a/R)}.$$

Substitute back and collect:

$$\boxed{p(r, t) = p_0 - \frac{3\rho\nu}{h^3} \left[(R^2 - r^2) - (R^2 - a^2) \frac{\ln(r/R)}{\ln(a/R)} \right] \dot{h}.} \quad (21)$$

(c) Lift force and effect of a hole

For $\dot{h} > 0$, $p < p_0$ throughout the film: atmospheric pressure on the top of the disk exceeds the film pressure below, so the fluid sucks the disk down and a lift force is needed.

Net upward fluid force:

$$F = \int_a^R [p_0 - p(r, t)] 2\pi r \, dr.$$

Set $a = 0$: the second bracket of (21) vanishes (since $(R^2 - a^2)/\ln(a/R) \rightarrow 0$):

$$p_0 - p(r, t) = \frac{3\rho\nu\dot{h}}{h^3} (R^2 - r^2).$$

Substitute:

$$F = \frac{6\pi\rho\nu\dot{h}}{h^3} \int_0^R (R^2 - r^2) r \, dr.$$

Evaluate the integral:

$$\int_0^R (R^2 - r^2)r \, dr = \frac{R^4}{2} - \frac{R^4}{4} = \frac{R^4}{4}.$$

Therefore

$$F = \frac{3\pi\rho\nu\dot{h}R^4}{2h^3}. \quad (22)$$

Insert numbers. Dynamic viscosity:

$$\mu = \rho\nu = 10^{-3} \text{ Pa s}.$$

Numerator:

$$3\pi\mu\dot{h}R^4 = 3\pi \times (10^{-3} \text{ Pa s}) \times (10^{-3} \text{ m s}^{-1}) \times (1 \text{ m}^4) = 3\pi \times 10^{-6} \text{ N m}^3.$$

Denominator:

$$2h^3 = 2 \times (5 \times 10^{-4} \text{ m})^3 = 2.5 \times 10^{-10} \text{ m}^3.$$

Divide:

$$F = \frac{3\pi \times 10^{-6}}{2.5 \times 10^{-10}} \text{ N} \approx 3.77 \times 10^4 \text{ N}.$$

$$F \approx 3.8 \times 10^4 \text{ N} \quad (\text{about } 4 \text{ tonnes-force}). \quad (23)$$

Effect of a small hole: $p(a) = p_0$ pins the pressure to atmospheric over the most strongly suctioned central region. Even small a removes the deepest part of the pressure deficit, dramatically reducing F — the same principle as a leak releasing a suction cup. The h^{-3} scaling otherwise causes the resistive force to diverge as the gap closes.

Question 4 — Sound waves: linear theory and nonlinear steepening

Problem. 1D adiabatic Euler with $p/p_0 = (\rho/\rho_0)^\gamma$, $\gamma = 7/5$. (a) Linearise about rest, derive the wave equation, show $c_{\text{ph}} = c_{\text{gr}} = \sqrt{dp/d\rho}$, and estimate c at standard atmosphere ($p_0 = 10^5 \text{ Pa}$, $\rho_0 = 1.2 \text{ kg m}^{-3}$). (b) With $dP = dp/(\rho c)$, derive the Riemann form, take $v_x = P$ to make the second equation trivial, interpret the first as a material derivative, sketch the evolution of $v_x = v_0 \sin(kx)$, and estimate the breaking time.

(a) Linear sound waves and sound speed

Perturb: $v_x = v'_x$, $\rho = \rho_0 + \rho'$, $p = p_0 + p'$, all primes first order. Linearise momentum:

$$\frac{\partial v'_x}{\partial t} + \frac{1}{\rho_0} \frac{\partial p'}{\partial x} = 0. \quad (24)$$

Linearise continuity:

$$\frac{\partial \rho'}{\partial t} + \rho_0 \frac{\partial v'_x}{\partial x} = 0. \quad (25)$$

Expand the adiabat to first order in ρ'/ρ_0 :

$$p_0 + p' = p_0 \left(1 + \frac{\rho'}{\rho_0} \right)^\gamma \approx p_0 \left(1 + \gamma \frac{\rho'}{\rho_0} \right).$$

Read off:

$$p' = \gamma \frac{p_0}{\rho_0} \rho' = c^2 \rho', \quad c^2 \equiv \left. \frac{\partial p}{\partial \rho} \right|_s = \gamma \frac{p_0}{\rho_0}.$$

Differentiate (25) in t :

$$\frac{\partial^2 \rho'}{\partial t^2} + \rho_0 \frac{\partial}{\partial x} \frac{\partial v'_x}{\partial t} = 0.$$

Substitute (24) and $p' = c^2 \rho'$:

$$\frac{\partial^2 p'}{\partial t^2} - c^2 \frac{\partial^2 p'}{\partial x^2} = 0. \quad (26)$$

Plane wave $p' \propto e^{i(kx - \omega t)}$ in (26): $\omega^2 = c^2 k^2$, hence

$$c_{\text{ph}} = \omega/k = c, \quad c_{\text{gr}} = d\omega/dk = c.$$

$$\boxed{c_{\text{ph}} = c_{\text{gr}} = c = \sqrt{\frac{dp}{d\rho}}.} \quad (27)$$

Insert standard atmosphere:

$$c^2 = \frac{7}{5} \cdot \frac{10^5}{1.2} \text{ m}^2 \text{ s}^{-2} = \frac{7 \times 10^5}{6} \text{ m}^2 \text{ s}^{-2}.$$

Square root:

$$\boxed{c \approx 341 \text{ m s}^{-1}.}$$

$$\boxed{c \approx 340 \text{ m s}^{-1}.} \quad (28)$$

(b) Riemann invariants, simple wave and shock formation

Define $dP = dp/(\rho c)$. For the adiabat $c^2 = \gamma p/\rho$, so $c \propto \rho^{(\gamma-1)/2}$ and $P(\rho)$ is single-valued. Rewrite $\rho^{-1} \partial_x p$:

$$\frac{1}{\rho} \frac{\partial p}{\partial x} = \frac{c^2}{\rho} \frac{\partial \rho}{\partial x} = c \frac{\partial P}{\partial x}.$$

Momentum becomes:

$$\frac{\partial v_x}{\partial t} + v_x \frac{\partial v_x}{\partial x} + c \frac{\partial P}{\partial x} = 0. \quad (29)$$

Divide continuity by ρ/c using $d\rho = (\rho/c) dP$:

$$\frac{\partial P}{\partial t} + v_x \frac{\partial P}{\partial x} + c \frac{\partial v_x}{\partial x} = 0. \quad (30)$$

Add (29)+(30):

$$\boxed{\left[\frac{\partial}{\partial t} + (v_x + c) \frac{\partial}{\partial x} \right] (v_x + P) = 0.} \quad (31)$$

Subtract:

$$\boxed{\left[\frac{\partial}{\partial t} + (v_x - c) \frac{\partial}{\partial x} \right] (v_x - P) = 0.} \quad (32)$$

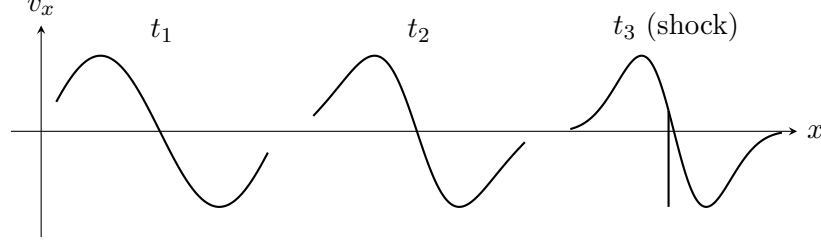
$R_{\pm} = v_x \pm P$ are conserved on characteristics $dx/dt = v_x \pm c$.

Take $v_x = P$ everywhere initially: $R_- = 0$, preserved by (32), so the second equation is trivial. Then (31) reduces to:

$$\frac{\partial v_x}{\partial t} + (v_x + c) \frac{\partial v_x}{\partial x} = 0. \quad (33)$$

This is $Dv_x/Dt = 0$ along $dx/dt = v_x + c$: v_x is constant on each characteristic, which is therefore a straight line of slope $1/(v_x + c)$. Larger v_x means a steeper line, so crests travel faster than troughs.

For $v_x(x, 0) = v_0 \sin(kx)$, positive- v_x regions advance at $c + v_x > c$ and negative- v_x regions lag at $c + v_x < c$. The leading face steepens, the trailing face flattens.



Estimate the breaking time. Two close characteristics carrying v_x and $v_x + \Delta v_x$ separated by Δx converge at relative speed Δv_x :

$$\tau \sim \frac{\Delta x}{\Delta v_x}.$$

Use $\Delta v_x \sim \Delta x \partial_x v_x$ at the steepest-descent point ($\partial_x v_x|_{\max} = -kv_0$):

$$\tau_{\text{break}} \sim \frac{1}{kv_0}.$$

The implicit-solution analysis gives prefactor unity:

$$\boxed{\tau_{\text{break}} = \frac{1}{kv_0}}. \tag{34}$$